

AI Weather API Microservice

01. EXECUTIVE SUMMARY

The **AI Weather API** is a cloud-native serverless solution that bridges standard meteorological parameters with Generative AI capabilities. Traditional weather payloads deliver strict, non-contextual numerical metrics. This system aggregates real-time metrics (temperature, humidity, pressure, and wind dynamics) from OpenWeatherMap, feeds the structured state into Google Gemini 2.5 Flash, and synthesizes an actionable, human-readable weather briefing on demand.

Serverless Core

Built entirely on AWS Lambda and HTTP API Gateway v2, offering high-concurrency autoscaling with zero baseline idle cost.

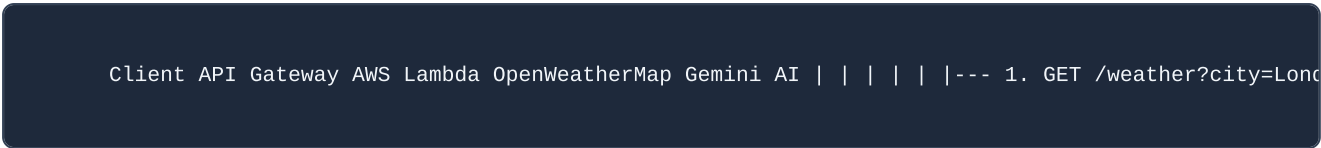
Generative AI Engine

Uses Google Gemini AI to deliver localized 2-sentence weather insights tailored to outdoor activities and clothing.

02. HIGH-LEVEL SYSTEM TOPOLOGY

+-----+ | CLIENT

03. SEQUENCE LIFECYCLE



04. INTERFACE CONTRACT SPECIFICATION

Endpoint: GET /weather

Parameter	Type	Required	Description
city	String	Yes	Target city for weather retrieval (e.g., London, Tokyo)

05. OPERATIONAL REQUIREMENTS & SLAS

Metric	SLA / Target	Mitigation Strategy
End-to-End Latency	< 1200 ms (p95)	Asynchronous connection pooling via httpx engine
Availability	99.9% Uptime	Multi-AZ deployment via AWS Serverless Infrastructure
Fault Tolerance	Graceful Degradation	Returns raw weather metrics if Gemini AI endpoint times out
Security	Secret Isolation	API Keys stored in AWS Secrets Manager / Encrypted Lambda Env